HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Designed and executed large-scale experiments to evaluate **LLM architectures** and **agentic workflows**, establishing clear hypotheses and reproducible results for AI platform strategy.
- Investigated and optimized **retrieval systems**, **embeddings**, and **context management** strategies to enhance the performance of AI applications.
- Contributed to the design and development of production-grade **AI infrastructure**, including **evaluation frameworks** and **model routing** capabilities.
- Assessed AI system performance, scalability, and reliability, identifying architectural opportunities for improvement in **large-scale experiments**.
- Collaborated with platform architects and engineering teams to translate experimental findings into practical, **scalable AI solutions**.
- Produced clear technical reports and experimental findings that informed platform strategy and engineering decisions for **emerging AI tools**.
- Containerized **Python-based AI services** using **Docker** and deployed them to a managed Kubernetes cluster, configuring basic resource limits for predictable rollouts.

SKILLS

AI/ML – Large Language Models (LLMs), Agentic AI Systems, Retrieval Systems, Embeddings, Context Management, Transformers, ML Fundamentals, Statistical Learning, Model Monitoring, Evaluation Frameworks, AI Platform Architecture

Programming – Python, Java, SQL

Frameworks/Libraries – LangChain, scikit-learn, pandas, NumPy, Spring Boot, FastAPI

Infrastructure – AWS (EC2, S3, Lambda, CloudWatch), Docker, Kubernetes, Linux

Tools – Git, Jenkins, Jira, Splunk, Dynatrace, Confluence

Methodologies – Agile, Scrum, CI/CD, Experiment Design, Technical Research

PROJECTS

AI Experimentation Framework for LLM Evaluation

- Developed a Python-based experimentation framework to systematically evaluate different **LLM prompting strategies** and **retrieval-augmented generation (RAG)** approaches.
- Implemented automated benchmarking for context window utilization and **response latency** across multiple foundation models, processing over 1,500 test queries.
- Designed and integrated a **vector database** (Pinecone) for efficient semantic search and context retrieval, reducing average retrieval time by 40%.

Agentic Workflow Orchestration Service

- Built a **FastAPI service** in Python to orchestrate multi-step agentic workflows, enabling AI agents to interact with external tools and APIs.
- Integrated with a suite of simulated enterprise tools, allowing agents to perform tasks like data retrieval and report generation, successfully completing ~95% of defined task sequences.
- Containerized the service using **Docker** and deployed it to an **AWS Lambda** environment for scalable execution of agent tasks.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)